

## 1 Research to be completed

- 1 Monte Carlo method for electronic ground state estimation
  - 1.1 Validity of and extensions for LEGS for larger molecules
  - 1.2 Treatment of multireferentiality
- 2 Potential energy extrapolation via coherent state transport
  - 2.1 How to best adiabatically deform coherent states?
  - 2.2 Can we retrieve PESs with good precision by transferring a sample of states into which an FCI solution was decomposed at one point in the parameter space?
  - 2.3 How does the method fare when encountering surface crossings (degeneracy)?
  - 2.4 Multireferentiality, cloning... and other potentially improving ideas
- 3 Theoretical analysis of Qubit states (a.k.a. ACPNCCS)
  - 3.1 Topology of the Qubit state manifold
  - 3.2 Coherence of Qubit states under static Hamiltonian evolution

## 2 Project risks / mitigation

The project, being partially theoretical and partially computational in nature, bears neither any risks to safety nor health or equipment hazards. There is no sensitive data collected.

## 3 Statement of confidence of submitting on time

I declare that I intend to submit my thesis on time and do not see any significant issues that may impede my progress. I am confident that I can uphold the deadlines given by the University.